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Establishing Information Transfer for Multilingual Intent Detection and Slot Filling

(/pdf?id=R7rjduAm6S)

ACL ARR 2025 February Submission2654 Authors

15 Feb 2025 (modified: 27 Mar 2025) ACL ARR 2025 February Submission February, Senior Area Chairs, Area Chairs, Reviewers, Authors Revisions (/revisions?id=R7rjduAm6S) CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/)

Abstract:
Multilingual spoken language understanding (SLU) involves intent detection (ID) and slot filling (SF) across multiple languages. The inherent linguistic diversity presents significant challenges in achieving performance comparable to traditional SLU. Recent studies have attempted to improve multilingual SLU performance by sharing multilingual encoders. However, these approaches have not directly established information flow between languages. To address this, we first demonstrate the feasibility of such information transfer and pinpoint the key challenges: prediction error mitigation and multilingual slot alignment. We then propose the INformation Transfer network (INT) to tackle these challenges. The gate unit in INT controls the information flow between languages, reducing the adverse impact of prediction errors on both ID and SF. Additionally, we reformulate SF as a span prediction problem and introduce a slot-matching attention mechanism to achieve slot alignment across languages. Experimental results on the MASSIVE and MASSIVE-UG datasets show that our model outperforms all baselines in overall accuracy across all languages, and demonstrates robust performance when different languages are used as the source.

Paper Type: Long
Research Area: Multilingualism and Cross-Lingual NLP
Research Area Keywords: Multilingualism and Cross-Lingual NLP
Languages Studied: Afrikaans (South Africa), Amharic (Ethiopia), Arabic (Saudi Arabia), Azerbaijani (Azerbaijan), Bengali (Bangladesh), Welsh (United Kingdom), Danish (Denmark), German (Germany), Greek (Greece), English (United States), Spanish (Spain), Persian (Iran), Finnish (Finland), French (France), Hebrew (Israel), Hindi (India), Hungarian (Hungary), Armenian (Armenia), Indonesian (Indonesia), Icelandic (Iceland), Italian (Italy), Japanese (Japan), Javanese (Indonesia), Georgian (Georgia), Khmer (Cambodia), Kannada (India), Korean (South Korea), Latvian (Latvia), Malayalam (India), Mongolian (Mongolia), Malay (Malaysia), Burmese (Myanmar), Norwegian Bokmål (Norway), Dutch (Netherlands), Polish (Poland), Portuguese (Portugal), Romanian (Romania), Russian (Russia), Slovenian (Slovenia), Albanian (Albania), Swedish (Sweden), Swahili (Kenya), Tamil (India), Telugu (India), Thai (Thailand), Tagalog (Philippines), Turkish (Turkey), Uyghur (China), Urdu (Pakistan), Vietnamese (Vietnam), Chinese (China), Chinese (Taiwan)
Reassignment Request Area Chair: This is not a resubmission
Reassignment Request Reviewers: This is not a resubmission
A1 Limitations Section: This paper has a limitations section.
A2 Potential Risks: N/A
B Use Or Create Scientific Artifacts: No
B1 Cite Creators Of Artifacts: Yes
B1 Elaboration: 1, 2, 3, 4, 5
B2 Discuss The License For Artifacts: N/A
B2 Elaboration: N/A
B3 Artifact Use Consistent With Intended Use: N/A
B3 Elaboration: N/A
B4 Data Contains Personally Identifying Info Or Offensive Content: N/A
B5 Documentation Of Artifacts: N/A
B5 Elaboration: N/A
B6 Statistics For Data: Yes
B6 Elaboration: 2
C Computational Experiments: Yes
C1 Model Size And Budget: Yes
C1 Elaboration: 4

C2 Experimental Setup And Hyperparameters: Yes
C2 Elaboration: 4
C3 Descriptive Statistics: Yes
C3 Elaboration: 4
C4 Parameters For Packages: Yes
C4 Elaboration: 4
D Human Subjects Including Annotators: No
D1 Instructions Given To Participants: N/A
D2 Recruitment And Payment: N/A
D3 Data Consent: N/A
D3 Elaboration: The dataset used in this paper is publicly available.
D4 Ethics Review Board Approval: N/A
D5 Characteristics Of Annotators: N/A
E Ai Assistants In Research Or Writing: No
E1 Information About Use Of Ai Assistants: N/A
E1 Elaboration: N/A
Author Submission Checklist: yes
Submission Number: 2654

Discussion (/forum?id=R7rjduAm6S#discussion)

Committee Members Chat (/forum?id=R7rjduAm6S#committee-chat)

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17 / 17 replies shown

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Official Review of Submission2654 by Reviewer edYX

Official Review by Reviewer edYX 02 Apr 2025, 05:35 (modified: 02 Apr 2025, 05:39)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer edYX

Revisions (/revisions?id=pHMtc9YWWS)

Paper Summary:

The paper presents the Information Transfer network (INT), a new model aimed at improving multilingual spoken language understanding, especially intent detection and slot filling. Unlike prior methods, INT explicitly transfers information between languages, addressing two main challenges, which are: reducing prediction errors and aligning multilingual slots.

INT utilises a gating mechanism to mitigate errors and a slot-matching attention module combined with span-based slot filling to handle multilingual differences effectively. The authors perform experiments on the MASSIVE52 dataset to show that INT significantly outperforms existing multilingual models like XLM-R and mT5.

Summary Of Strengths:

1. The paper clearly introduces an innovative methodology for direct information transfer between languages for multilingual SLU. This differentiates significantly from prior research and fills a critical gap.
2. The experimental evaluation is robust, covering a wide range of baselines including mPTMs and large language models (LLMs).
3. The paper is well-organized, logically structured, and clearly written.

Summary Of Weaknesses:

1. While the paper acknowledges cases where performance may degrade, it does not provide detailed qualitative error analysis. Understanding specific scenarios where INT struggles or even negatively impacts performance could provide

deeper insights into model limitations.
2. Although the authors briefly address the impact of the source language performance, a deeper exploration of how different source languages with varying linguistic distances affect performance would add important insights.

Comments Suggestions And Typos:
1. Include a detailed qualitative and quantitative error analysis, identifying particular intents or slots where INT underperforms compared to baselines.
2. Validate the proposed model on additional multilingual SLU datasets or real-world data scenarios beyond MASSIVE52.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.
Soundness: 4 = Strong: This study provides sufficient support for all of its claims. Some extra experiments could be nice, but not essential.
Excitement: 3.5
Overall Assessment: 4.0 = Conference: I think this paper could be accepted to an *ACL conference.
Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No
Reproducibility: 5 = They could easily reproduce the results.
Datasets: 1 = No usable datasets submitted.
Software: 5 = Enabling: The newly released software should affect other people's choice of research or development projects to undertake.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Add: Official Comment

Official Comment by Authors

Official Comment by Authors 03 Apr 2025, 11:00

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer edYX

Comment:
Dear Reviewer edYX,

Thank you for your valuable feedback. Below are our responses to your comments:

Q1: While the paper acknowledges cases where performance may degrade, it does not provide detailed qualitative error analysis. Understanding specific scenarios where INT struggles or even negatively impacts performance could provide deeper insights into model limitations.

R1: We commit to supplementing Section 4.7 with error case examples for qualitative analysis.

As shown in Section 4.7, a small subset of categories exhibited decreased rather than improved accuracy. Without information transfer, the utterance intent label 'qa_factoid' is correctly assigned. However, when information transfer is implemented, incorrect prediction of English utterance intent labels (e.g., 'qa_definition') introduces cascading interference. Although our gating mechanism partially mitigates the impact of source-language prediction errors, some utterances remain affected when errors occur.

We will augment the manuscript with both correctly transferred examples and cases affected by prediction interference, accompanied by detailed analysis.

Q2: Although the authors briefly address the impact of the source language performance, a deeper exploration of how different source languages with varying linguistic distances affect performance would add important insights.

R2: We commit to supplementing the analysis in Section 4.6 to examine the effects of establishing information transfer between linguistically similar and distant languages.

When using English (en-US) to establish information transfer, closely related Dutch (nl-NL) shows an overall accuracy improvement of 4.95%, while more distant Chinese (zh-CN) achieves a greater improvement of 10.93%. This discrepancy stems from XLM-R's baseline performance differences between nl-NL and zh-CN - nl-NL's original accuracy was already 5.68% higher than zh-CN, hence zh-CN demonstrates more significant gains.

Two primary factors influence INT's performance across languages:

1. The language's pre-transfer performance - languages with poorer initial performance tend to show greater improvement;
2. The source language's performance, which exhibits a direct positive correlation with INT's results.

From the study's inception, we aimed to develop a solution that establishes information transfer purely based on semantics, thereby minimizing impacts from language families, linguistic distances, or encoding differences, and maximizing abstract semantic information extraction. Our model has, to some extent, achieved this objective and has minimized the influence of language-specific characteristics.

We commit to incorporating this comprehensive analysis into the manuscript.

Q3: Validate the proposed model on additional multilingual SLU datasets or real-world data scenarios beyond MASSIVE52.

R3: MASSIVE52 is the multilingual SLU dataset formed by combining MASSIVE and MASSIVE(UG), covering 52 languages. It represents the publicly available multilingual SLU dataset with the largest number of languages we could find. We will continue to monitor advancements in multilingual SLU research and consistently evaluate our proposed INT framework on new multilingual SLU datasets.

Once again, thank you for your valuable feedback.

Add: Official Comment

Official Review of Submission2654 by Reviewer kv29

Official Review by Reviewer kv29 27 Mar 2025, 06:17 (modified: 28 Mar 2025, 17:23)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer kv29

Revisions (/revisions?id=VoOqCYyYH1)

Paper Summary:
The paper introduces the INformation Transfer network (INT), a model designed to transfer information from one source language to multiple target languages for multilingual spoken language understanding (SLU), specifically intent detection and slot filling. Existing multilingual SLU models don't explicitly transfer information across languages, limiting cross-lingual effectiveness. The authors identify two key challenges: (1) prediction errors - transferring incorrect information can hurt performance, and (2) slot alignment - slot spans often differ in position or length across languages. To address these, INT transfers intent and slot information with a trainable gate that controls how much is transferred, and reformulates slot filling as a span prediction task to better handle structural differences. It also introduces a slot-matching attention mechanism to align semantically similar spans across languages. Evaluated on the MASSIVE52 dataset against several baselines (XLM-R, mT5, three mPTMs, three LLMs), INT outperforms all models across key metrics. Ablation studies further show that the gates and slot-matching components are crucial to its success.

Summary Of Strengths:
The paper introduces a novel approach to multilingual spoken language understanding by directly transferring intent and slot information across languages, moving beyond encoder-sharing methods. Key innovations include a trainable gating mechanism to regulate transfer and a span-based slot filling strategy with slot-matching attention for better cross-lingual alignment. The model is thoroughly evaluated on the MASSIVE52 dataset, outperforming strong baselines like XLM-R, mT5, three mPTMs, and three LLMs. Extensive ablation studies further validate the contribution of each component, highlighting the robustness and effectiveness of the proposed method.

Summary Of Weaknesses:
1. The evaluation is heavily focused on quantitative results; incorporating some qualitative analysis, such as illustrative examples or error analysis, would provide more insight into how and why the model succeeds or fails. This would be especially helpful given the authors' own observation that performance decreases in a small subset of categories.
2. The reliance on parallel data between source and target languages during training limits the feasibility of applying the method in real-world, low-resource scenarios.
3. The paper could benefit from some format improvements, such as more descriptive figure and table captions and relocating the related work section to earlier in the paper for better contextual framing.

Comments Suggestions And Typos:

1. Section 3 (p.3) heading typo: “Mehtod” should be corrected to “Method.”
2. Figure 4 caption (p.4) is too minimal—consider expanding it to explain the components and flow of the model. Clearer figure/table captions throughout would improve readability.
3. Subsection title on p.7: “Complementarity between intent gate gate” contains a duplicate word (gate).
4. The Related Work section appears near the end of the paper, but moving it closer to the introduction would better contextualize the contribution.

Confidence: 2 = Willing to defend my evaluation, but it is fairly likely that I missed some details, didn't understand some central points, or can't be sure about the novelty of the work.
Soundness: 4 = Strong: This study provides sufficient support for all of its claims. Some extra experiments could be nice, but not essential.
Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.
Overall Assessment: 4.0 = Conference: I think this paper could be accepted to an *ACL conference.
Best Paper Justification:
N/A

Limitations And Societal Impact:
Yes, they've discussed the limitations.

Ethical Concerns:
There are no concerns with this submission.

Needs Ethics Review: No
Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.
Datasets: 1 = No usable datasets submitted.
Software: 1 = No usable software released.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Add: Official Comment

Official Comment by Authors

Official Comment by Authors 30 Mar 2025, 19:59

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer kv29

Comment:
Dear Reviewer kv29,

Thank you for your thoughtful feedback. Below are our responses to your questions:

Q1: The evaluation is heavily focused on quantitative results; incorporating some qualitative analysis, such as illustrative examples or error analysis, would provide more insight into how and why the model succeeds or fails. This would be especially helpful given the authors’ own observation that performance decreases in a small subset of categories.

R1: We commit to supplementing the manuscript with case studies and error analysis while strictly adhering to conference guidelines.

When the information transfer is not established, the utterance is correctly classified with the intent label "qa_factoid". However, when the transfer is established, prediction errors in the English utterance intent labeling (incorrect "qa_definition") trigger cascading error propagation. Although our gating mechanism partially mitigates source-language prediction errors, interference persists for a subset of utterances when such errors occur.

We will add a dedicated subsection to comprehensively illustrate: (1) Exemplary cases of successful information transfer. (2) Representative examples of prediction interference caused by error propagation.

Q2: The reliance on parallel data between source and target languages during training limits the feasibility of applying the method in real-world, low-resource scenarios.

R2: Indeed, our current work focuses exclusively on multilingual spoken language understanding tasks. We will continue to refine and improve the proposed INT method, with the goal of extending it to cross-lingual spoken language understanding tasks in future research.

Q3: The paper could benefit from some format improvements, such as more descriptive figure and table captions and relocating the related work section to earlier in the paper for better contextual framing.

R3: We will optimize the table titles and formatting to improve the manuscript's readability.

Q4: Comments Suggestions And Typos.

R4: We will carefully correct all spelling errors, refine the title descriptions, and optimize the manuscript formatting to improve overall clarity.

Once again, thank you for your valuable suggestions.

Add: Official Comment

Replying to Official Comment by Authors

Official Comment by Reviewer kv29

Official Comment by Reviewer kv29 31 Mar 2025, 00:28

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer kv29

Comment:
Thank you for the point-by-point response. I look forward to the revised version.

Add: Official Comment

Reviewer Checklist of Submission2654 by Reviewer kv29

Reviewer Checklist by Reviewer kv29 27 Mar 2025, 01:53

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted

Appropriateness: Yes
Formatting: Yes
Length: Yes
Anonymity: Yes
Limitations: Yes
Responsible Checklist: Yes
Potential Violation Justification: There are no violations with this submission
Need Ethics Review: No
Ethics Review Justification: N/A (I answered no to the previous question)

Add: Official Comment

Official Review of Submission2654 by Reviewer 6SCa

(https://openreview.net/forum?id=R7rjduAm6S&noteId=1SA9PDHeer) Edit

Official Review by Reviewer 6SCa (Peijie Huang (/profile?id=~Peijie_Huang1))

23 Mar 2025, 16:53 (modified: 28 Mar 2025, 17:23)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 6SCa

Revisions (/revisions?id=1SA9PDHeer)

Paper Summary:

This paper focuses on improving multilingual spoken language understanding (SLU) by addressing intent detection (ID) and slot filling (SF) challenges. It introduces the INformation Transfer Network (INT), which facilitates direct information transfer between languages. The model uses a gating mechanism to mitigate prediction errors and slot-matching attention to align slots across different languages. Experiments on MASSIVE and MASSIVE-UG datasets show that INT outperforms existing methods in accuracy across multiple languages.

Summary Of Strengths:

1. Introduces an explicit information transfer mechanism between languages.
2. Effectively mitigates prediction errors using a trainable gating unit.
3. Reformulates slot filling as span prediction, improving multilingual alignment.
4. Outperforms baseline models on established multilingual SLU datasets.

Summary Of Weaknesses:

1. Some typos (e.g., Line 70 "?").
2. The proposed method is not novel in the SLU community, as many works have used the "span" concept to tackle token-level tagging tasks. The author seems to have merely transferred the concept from another domain.
3. Why not use open-source LLMs like Llama3-8B as baselines with SFT? It seems less convincing in the current research domain to rely solely on prompt engineering LLMs.
4. Some challenging like Error Mitigation or Alignment are very common in the whole SLU field, so I don't think it worth servering as the part of motivation.

Comments Suggestions And Typos:
See weaknesses section.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.
Soundness: 2.5
Excitement: 2 = Potentially Interesting: this paper does not resonate with me, but it might with others in the *ACL community.
Overall Assessment: 2.5 = Borderline Findings
Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No
Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.
Datasets: 1 = No usable datasets submitted.
Software: 1 = No usable software released.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.
Secondary Reviewer: Shangjian Yin (/profile?id=~Shangjian_Yin1)

Add: Official Comment

Official Comment by Authors

Official Comment by Authors 30 Mar 2025, 20:11

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 6SCa

Comment:
Dear Reviewer 6SCa,

Thank you for your constructive feedback.

Q1: The proposed method is not novel in the SLU community, as many works have used the "span" concept to tackle token-level tagging tasks. The author seems to have merely transferred the concept from another domain.

R1: Our research motivation lies in establishing information transfer between languages, where "span" merely serves as a decoding method rather than the core objective. As discussed in Section 4.5, the role of *span* is solely for slot information aggregation. Even without "span", our proposed INT still outperforms the baseline model by

5.11% in overall accuracy (Table 1 and Figure 5). The superior performance of INT primarily stems from its explicit modeling of information transfer between languages.

Q2 : Why not use open-source LLMs like Llama3-8B as baselines with SFT? It seems less convincing in the current research domain to rely solely on prompt engineering LLMs.

R2: While Llama3-8B supports 8 languages, the dataset encompasses 52 languages, making it unsuitable as a baseline for this task. We therefore selected BLOOMZ (46 languages) and GLM-4 (26 languages) as baselines due to their broader language coverage. For our LLM experiments, we employed not only prompt engineering but also LoRA-based fine-tuning, as detailed in Section 4.2 of our manuscript.

Q3: Some challenging like Error Mitigation or Alignment are very common in the whole SLU field, so I don't think it worth servering as the part of motivation.

R3: The baseline model exhibits inconsistent performance across languages, demonstrating strong results in certain languages while delivering mediocre performance in others. The primary objective of this work is to uniformly improve spoken language understanding (SLU) performance across all languages through information transfer. Error mitigation and alignment challenges are potential issues that may arise in this process, but they are not our primary research motivation. For details, please refer to Section 1.

Q4: Comments Suggestions And Typos.

R4: We will carefully correct all spelling errors in the manuscript.

Once again, thank you for your valuable review.

Add: Official Comment

Replying to Official Comment by Authors

Official Comment by Reviewer 6SCa

Edit

Official Comment by Reviewer 6SCa (Peijie Huang (/profile?id=~Peijie_Huang1)) 02 Apr 2025, 01:31

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 6SCa

Comment:
Thank you for your rebuttal, however, the clafications of motivation and method are not convincing to me. I suggest authors to conduct expriments on advanced LLM like LLama3.1 and futher polish the motivation paragraph. I tend to maintain my score.

Add: Official Comment

Replying to Official Comment by Reviewer 6SCa

Official Comment by Authors

(https://openreview.net/forum?id=R7rjduAm6S&noteId=oPhfsG5Vi1)

Official Comment by Authors 03 Apr 2025, 11:40

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 6SCa

Comment:
Dear Reviewer 6SCa,

Thank you for getting back to us in a timely manner.

Q4: Thank you for your rebuttal, however, the clafications of motivation and method are not convincing to me. I suggest authors to conduct expriments on advanced LLM like LLama3.1 and futher polish the motivation paragraph. I tend to maintain my score.

R4:

1. The research motivation of our manuscript is to explicitly establish information transfer in multilingual SLU tasks. To our knowledge, such an approach has not been previously explored in existing multilingual SLU studies (Second paragraph of the introduction).
2. The methodological innovations have been elaborated in R1 - even without using span representations, our approach still achieves a 5.11% improvement in overall accuracy compared to baselines.

May we understand the reasons for remaining reservations?

We agree your concerns regarding the Llama3.1-8B experiments and are now reporting the experimental results.

Model	Intent (ACC)	Slot (F1)	Overall (ACC)
GPT-4	70.72	41.90	29.20
BLOOMz	75.60	61.54	50.95
Llama3.1	84.43	71.02	61.82
GLM-4	84.24	71.90	62.03
INT(Ours)	88.21	80.75	70.65

As can be seen, our proposed INT outperforms Llama3.1-8B when using XLM-R as the encoder on the MASSIVE52 dataset. We commit to supplementing the manuscript with the experimental results and corresponding analysis for Llama3.1-8B.

We hope our current response has adequately addressed your concerns, and we wish to express our sincere gratitude once again for your valuable feedback.

Add: Official Comment

Replying to Official Comment by Authors

Official Comment by Authors

Official Comment by Authors 04 Apr 2025, 15:58

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer 6SCa

Comment:
Dear 6SCa,

As the discussion period nears its conclusion, we wanted to follow up to ensure that our responses have sufficiently addressed your concerns. Please feel free to let us know if there is anything else we can clarify. We sincerely appreciate the time and effort you have dedicated to reviewing our work and look forward to your feedback.

Add: Official Comment

Official Review of Submission2654 by Reviewer BofZ

Official Review by Reviewer BofZ 21 Mar 2025, 19:17 (modified: 28 Mar 2025, 17:23)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer BofZ

Revisions (/revisions?id=90niRzbHly)

Paper Summary:
This paper propose the INformation Transfer network (INT), which facilitates information transfer between languages to improve performance. The key challenges identified are prediction error mitigation and multilingual slot alignment. The INT model introduces a gating unit to control information flow and mitigate errors, and reformulates SF as a span prediction problem to handle varying slot lengths and positions across languages. Additionally, a slot-matching attention mechanism is proposed to align slots across languages. The model is evaluated on the MASSIVE and MASSIVE-UG datasets, demonstrating superior performance over baseline models across multiple languages.

Summary Of Strengths:
Compared to existing multilingual SLU works, this paper better bridges the gap between different languages by establishing direct information transfer. On the one hand, the proposed INformation Transfer network (INT) facilitates knowledge transfer across diverse languages, allowing them to share intent and slot information effectively. On the other hand, the paper reformulates slot filling as a span prediction problem, which is more robust to the inconsistencies in slot lengths and positions across languages.

Summary Of Weaknesses:

Overall, the novelty of this work is limited. The key components of the proposed INformation Transfer network (INT), such as the gating mechanism for prediction error mitigation and the slot-matching attention mechanism for multilingual slot alignment, build upon well-established techniques in multilingual SLU and cross-lingual transfer learning.

Comments Suggestions And Typos:
The paper could benefit from a deeper exploration of specific multilingual SLU challenges, such as handling morphological complexity and code-switching, and evaluating the model's robustness in low-resource language settings. Comparisons with recent advancements like adapter-based fine-tuning and contrastive learning are missing.

Confidence: 5 = Positive that my evaluation is correct. I read the paper very carefully and am familiar with related work.
Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.
Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.
Overall Assessment: 2.5 = Borderline Findings
Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No
Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.
Datasets: 1 = No usable datasets submitted.
Software: 3 = Potentially useful: Someone might find the new software useful for their work.
Knowledge Of Or Educated Guess At Author Identity: No
Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources
Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources
Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Add: Official Comment

Official Comment by Authors

Official Comment by Authors 30 Mar 2025, 20:40

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer BofZ

Comment:
Dear Reviewer BofZ,

Thank you for your valuable feedback. Below are our responses to your comments:

Q1: Overall, the novelty of this work is limited. The key components of the proposed Information Transfer network (INT), such as the gating mechanism for prediction error mitigation and the slot-matching attention mechanism for multilingual slot alignment, build upon well-established techniques in multilingual SLU and cross-lingual transfer learning.

R1: We agree that the proposed INT draws inspiration from existing work in SLU and cross-lingual transfer learning. However, our primary contributions lie in:

1. Novel Exploration: The direct establishment of information transfer between languages in multilingual SLU, which has not been systematically investigated in previous work (as detailed in the second paragraph of the Introduction);
2. Empirical Validation: The proposed INT achieves an average improvement of 5.69% in overall accuracy across all languages in MASSIVE52, demonstrating consistent enhancement regardless of the encoder.

While we recognize that individual components are not entirely novel in isolation, they collectively address the critical challenge of establishing information transfer in multilingual SLU, thereby providing a new solution framework for this task. We will refine the manuscript to better highlight the significance of our research.

Q2: The paper could benefit from a deeper exploration of specific multilingual SLU challenges, such as handling morphological complexity and code-switching, and evaluating the model's robustness in low-resource language settings.

R2: From the outset of our research, we aimed to develop an information transfer method that relies on semantic similarity for information transfer, rather than being limited to morphologically and typologically similar languages. This approach maximizes the utilization of abstract semantic knowledge within languages while minimizing the impact of morphological and typological differences.

Prior studies have demonstrated that establishing guidance between morphologically and typologically proximate languages is easier. In contrast, our proposed INT operates purely on a semantic basis, making it less susceptible to variations in morphology and typology.

As detailed in Section 4.6 of our manuscript, we evaluate INT's performance across different source languages (high-resource English and low-resource Japanese). Notably, even when using Japanese as the source language—a language with fewer resources—our model consistently outperforms the baseline.

Q3: Comparisons with recent advancements like adapter-based fine-tuning and contrastive learning are missing.

R3: The rationale for not adopting GL-CLEF, HC2L, or REPE as baselines lies in their reliance on code-switching to achieve cross-lingual alignment through syntactic reordering—an approach divergent from the multilingual SLU setup. We commit to revising the manuscript to clarify their contributions and applicable scope, while explaining our rationale for excluding them from the baselines.

Furthermore, we observed that adapter-based fine-tuning remains under-explored in SLU research. Consequently, the baseline employs LoRA-based methods, implemented as described in [1].

To further increase the diversity and challenge of our study, we supplemented the original 51 languages in MASSIVE (2023) with the newly available Uyghur language version (2024). The expanded language coverage led to our decision not to adopt [2] as a baseline.

[1] Do Large Language Model Understand Multi-Intent Spoken Language?

[2] HIT-SCIR at MMNLU-22: Consistency Regularization for Multilingual Spoken Language Understanding

Once again, thank you for your valuable feedback.

Add: Official Comment

Replying to Official Comment by Authors

Official Comment by Authors

Official Comment by Authors 04 Apr 2025, 15:59

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer BofZ

Comment:

Dear BofZ,

As the discussion period nears its conclusion, we wanted to follow up to ensure that our responses have sufficiently addressed your concerns. Please feel free to let us know if there is anything else we can clarify. We sincerely appreciate the time and effort you have dedicated to reviewing our work and look forward to your feedback.

Add: Official Comment

Reviewer Checklist of Submission2654 by Reviewer 6SCa

Edit

Reviewer Checklist by Reviewer 6SCa (Peijie Huang (/profile?id=~Peijie_Huang1)) 19 Mar 2025, 02:06

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted

Appropriateness: Yes
Formatting: Yes
Length: Yes
Anonymity: Yes

Limitations: Yes
Responsible Checklist: Yes
Potential Violation Justification: There are no violations with this submission
Need Ethics Review: No
Ethics Review Justification: N/A (I answered no to the previous question)

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Reviewer Checklist of Submission2654 by Reviewer 1ss7

Reviewer Checklist by Reviewer 1ss7 14 Mar 2025, 16:47
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted

Appropriateness: Yes
Formatting: Yes
Length: Yes
Anonymity: Yes
Limitations: Yes
Responsible Checklist: Yes
Potential Violation Justification: There are no violations with this submission
Need Ethics Review: No
Ethics Review Justification: N/A (I answered no to the previous question)

Add: Official Comment

Action Editor Checklist by Area Chair gmF2

Action Editor Checklist by Area Chair gmF2 (Alan Ramponi (/profile?id=~Alan_Ramponi1)) 02 Mar 2025, 00:26
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted

Appropriateness: Yes
Formatting: Yes
Length: Yes
Anonymity: Yes
Limitations: Yes
Responsible Checklist: Yes
Potential Violation Justification: There are no violations with this submission
Number Of Assignments: No
Diversity: Yes
Need Ethics Review: No
Ethics Review Justification: N/A (I answered no to the previous question)
Resubmission: This paper is a not a resubmission
Resubmission Reassignments: This paper is a not a resubmission

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